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Self-Confirming Equilibrium, Learning, and Observational Equivalence under Incomplete Information

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Introduction

Self-confirming equilibrium was introduced by Fudenberg and Levine as a weaker alternative to Nash equilibrium, meant to capture situations where players learn only from what they actually observe. It matters especially in repeated play, incomplete information, and cases of observational equivalence, where wrong off-path beliefs may never be disproved. (6 sources)

Self-confirming equilibrium (SCE) is one of the main attempts to relax Nash equilibrium when players do not fully learn or verify everything that could happen in a game. In surveys of equilibrium notions for incomplete information, it is listed alongside Bayesian equilibrium, correlated equilibrium, rationalizability, and subjective equilibrium, which helps place it as part of a broader response to limits of the standard Nash framework under uncertainty and partial feedback. (Wiszniewska-Matyszkiel, 2015) (Fudenberg et al., 1993) (Wiszniewska-Matyszkiel, 2016) (Wiszniewska-Matyszkiel, 2017)

The key idea, in the notable original formulation, is that each player chooses a best response to their beliefs, and those beliefs only need to be correct on the path of play that is actually observed. If the same outcome repeats, players may never see evidence against mistaken beliefs about actions or contingencies that never occur. This makes SCE a natural bridge between equilibrium and learning: it asks what patterns of play can persist when agents optimize using beliefs that are confirmed by experience, but only by the experience they in fact get. (Fudenberg et al., 1993) (Pieroth et al., 2025)

This perspective becomes especially useful for the user's themes of incomplete information and observational equivalence. Later work stresses that, in environments with possibly incomplete information and private values, the SCE set at a given state can be independent of players' interactive knowledge of payoff functions, which points to a central feature of the concept: what matters is not full mutual knowledge, but whether observed play leaves beliefs unrefuted. That is exactly why observational equivalence matters in SCE—different hidden models of the game can support the same observed behavior, so learning may stop short of Nash-level correctness. (Battigalli et al., 2024) (Fudenberg et al., 1993)

Evidence

(Wiszniewska-Matyszkiel, 2015)

Belief distorted Nash equilibria: introduction of a new kind of equilibrium in dynamic games with distorted information

"There are many attempts to extend the concept of Nash equilibria to make it work in the case of incomplete information: among others, Bayesian equilibria introduced by (Harsanyi, 1968), correlated equilibria introduced by (Aumann, 1974)(Rationality et al., 1987), -rationalizability introduced by (Battigalli et al., 2003), self-confirming equilibria introduced by (Fudenberg et al., 1993) and subjective equilibria introduced by Kalai andLehrer (1993, 1995)."

(Fudenberg et al., 1993)

Self-confirming equilibrium

"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts."

(Wiszniewska-Matyszkiel, 2016)

Redefinition of Belief Distorted Nash Equilibria for the Environment of Dynamic Games with Probabilistic Beliefs

"self-confirming equilibria by Fudenberg and Levine (Fudenberg et al., 1993)"

(Wiszniewska-Matyszkiel, 2017)

Redefinition of Belief Distorted Nash Equilibria for the Environment of Dynamic Games with Probabilistic Beliefs

"self-confirming equilibria by Fudenberg and Levine (Fudenberg et al., 1993)"

(Pieroth et al., 2025)

Algorithmic Predation: Equilibrium Analysis in Dynamic Oligopolies with Smooth Market Sharing

"Equilibrium learning offers an alternative numerical approach to finding equilibrium. It explores how equilibrium can emerge from agents that maximize their payoff while competing with each other [Fudenberg and Levine, 1999]."

(Battigalli et al., 2024)

Sophisticated reasoning, learning, and equilibrium in repeated games with imperfect feedback

"Indeed, in an environment with possibly incomplete information and private values, the SCE set at any given state of nature is independent of players' interactive knowledge of the profile of payoff functions"

".Fudenberg and Levine (1993) coined the term "self-confirming equilibrium.""

Core definition of self-confirming equilibrium

In Fudenberg and Levine's formulation, a self-confirming equilibrium requires two things: each player best responds to their beliefs, and those beliefs are right about what is actually observed in equilibrium. Beliefs about events or information sets that never occur can still be wrong, which is exactly why SCE is weaker than Nash equilibrium. (3 sources)

In the notable Fudenberg-Levine definition, a self-confirming equilibrium has a simple core structure: players choose strategies that maximize expected utility given their beliefs, and those beliefs must be confirmed by the realized path of play. Put differently, a player's conjecture about others only needs to fit what he actually sees, not every contingency in the whole game. (Fudenberg et al., 1993) (Sota et al., 2013)

This means the concept has two parts. First, **optimality**: each player's strategy is a best response to his or her subjective view of opponents' play. Second, **on-path correctness**: beliefs must be correct along the equilibrium path, meaning along the sequence of actions and signals that are actually generated. If the same pattern of play repeats, no one sees evidence that refutes these beliefs. (Fudenberg et al., 1993) (Macault et al., 2020)

The sharp difference from Nash equilibrium is that SCE does **not** require globally correct beliefs. A player can be wrong about what others would do at information sets that are never reached, or can hold views that are not fully consistent with rational play elsewhere, as long as observed play never contradicts them. That is the basic source of the gap between self-confirming and Nash equilibrium. (Fudenberg et al., 1993) (Macault et al., 2020)

A further useful point in the original subjective-game formulation is that beliefs can be fairly rich: they need not treat opponents' actions as independent, but can allow perceived positive or negative correlation across opponents' strategies. So the definition is not just about point beliefs over isolated actions; it allows players to best respond to broader subjective models of how others behave, provided those models are not contradicted by what is observed. (Sota et al., 2013)

Evidence

(Fudenberg et al., 1993)

Self-confirming equilibrium

> *"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts."*

(Sota et al., 2013)

How Albanian Private Universities can use Game Theory for Optimization of Scholarship Offers

> *"Fundenberg and Lebine (1993) introduced the notion of "self-confirming equilibrium" for finite subjective games. Each player in such a game chooses a subjective strategy to maximize his/her expected utility function, given his/her subjective beliefs about opponents' strategies. These beliefs allow the possibility that the opponents' strategies are positively (or negatively) correlated."*

(Macault et al., 2020)

Social learning in nonatomic routing games

> *"(Fudenberg et al., 1993) introduced the concept of self-confirming equilibrium, where players' beliefs about other players' actions are required to be correct only on the equilibrium path. As agents play best responses to their beliefs, these may differ substantially from the truth, as long as nothing contradicts the beliefs. Hence, self-confirming equilibria and Nash equilibria of a game need not be the same. Given the sequential nature of the repeated game, self-confirming equilibria may allow players to hold beliefs on other agents that may be inconsistent with rational behavior. Battigalli and Guaitoli (1997) refined self-confirming equilibrium by proposing the concept of conjectural equilibrium for extensive form games of incomplete information without a common prior."*

Learning foundations and steady-state interpretation

Fudenberg and Levine's main learning idea is that self-confirming equilibrium can describe a steady state of play when people update only from the feedback they actually get. If the same behavior keeps generating the same observations, beliefs can stabilize even when they are still wrong about unobserved contingencies. (11 sources)

A central use of self-confirming equilibrium is as a **learning foundation**: instead of taking equilibrium as a static consistency condition, the idea is to ask what outcomes can persist after repeated play and belief updating. Later work regularly describes the Fudenberg-Levine program in exactly these terms, as part of the literature that uses explicit learning models to justify Nash and self-confirming equilibrium. (Esponda et al., 2014) (Esponda et al., 2015)

The steady-state reading is especially important. A common summary is that a steady state of a belief-based learning process should be a self-confirming equilibrium, because players best respond to beliefs that fit the feedback generated by the play that keeps recurring. This is how later papers frame the original contribution when discussing repeated play in extensive-form games and other dynamic settings. (Dekel et al., 2004) (Fudenberg et al., 1993) (Jehiel et al., 2023)

What keeps learning from reaching Nash is **endogenous information acquisition**. Players only see consequences of the actions they choose and the paths that are reached, so some false beliefs may never be tested. Recent papers make this point by noting that fixed points of learning dynamics can differ from complete-information equilibrium precisely because wrong estimates about unobserved outcomes are never corrected; beliefs can even concentrate on different parameter values that all induce the same observed payoff distribution at the fixed point. (Wu et al., 2021) (Fudenberg et al., 1993) This same logic is now treated

as part of a broader phenomenon of incomplete learning under endogenous observation, with self-confirming equilibrium as one of the benchmark cases. (Wu et al., 2022) (Fudenberg et al., 1993)

This learning interpretation also shaped later refinements and extensions. Some authors explicitly say their concepts could be grounded by adding suitable assumptions to the steady-state learning model of Fudenberg and Levine, even when they do not model the full dynamics themselves. (Dekel et al., 1999) Others connect self-confirming equilibrium to patient Bayesian learning, where agents think they face a steady-state distribution of play, and then use experimentation and long-run feedback to sharpen which off-path beliefs can survive. (Fudenberg et al., 2017) (Fudenberg et al._1, 2017) (Fudenberg et al., 2006)

So, in the Fudenberg-Levine tradition, SCE is best understood not just as a weaker equilibrium notion, but as the **natural long-run outcome of learning with limited feedback**: play settles down when current beliefs keep being confirmed by observed data, even if deeper counterfactual beliefs remain wrong. (Esponda et al., 2014) (Dekel et al., 2004) (Fudenberg et al., 1993)

Evidence

(Esponda et al., 2014)

Berk-Nash Equilibrium: A Framework for Modeling Agents with Misspecified Models

> *"This literature studies explicit learning models in order to justify Nash and self-confirming equilibrium (e.g., Fudenberg and Kreps (1988), Fudenberg and Kreps (1993), Fudenberg and Kreps (1995), Fudenberg and Levine (1993b), Kalai and Lehrer (1993))"*
>
> *".Our framework unifies both standard solution concepts (Nash and self-confirming equilibrium) and a recent literature on bounded rationality and misspecified learning"*
>
> *".and D.K. Levine, "Self-confirming equilibrium," Econometrica, 1993, pp. 523-545."*

(Esponda et al., 2015)

Equilibrium in misspecified Markov decision processes

> *"More generally, the paper is related to the literature that provides learning foundations for equilibrium concepts, such as Nash or self-confirming equilibrium (see Fudenberg and Levine 1998 for a survey). In contrast to this literature, we consider Markov decision problems and allow for misspecified models."*

(Dekel et al., 2004)

Learning to play Bayesian games

> *"Fudenberg andKreps [1988] 1995] and (Fudenberg et al., 1993) examined learning, steady states, and self-confirming equilibria in games with non-trivial extensive forms"*
>
> *".We will not formally model the dynamics of learning, but will appeal informally to the idea that a steady state of a belief-based learning process must be a self-confirming equilibrium (Fudenberg et al., 1993)). Thus, our focus is on how the information that players observe at the end of each round of play determines the set of self-confirming equilibria, and how these equilibria relate to the Nash equilibria of the game."*

(Fudenberg et al., 1993)

Self-confirming equilibrium

> *"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts."*

(Jehiel et al., 2023)

The analogical foundations of cooperation

> *"Fudenberg andLevine's 1993a, 1993b self-confirming equilibrium similarly assumes that players play best responses to beliefs consistent with frequencies of play in a record, with the leeway in forming out-of-equilibrium beliefs playing an important role."*

(Wu et al., 2021)

Multi-agent Bayesian Learning with Best Response Dynamics: Convergence and Stability

"The notion of fixed point in our learning dynamics is similar to the self-confirming equilibrium introduced in (Fudenberg et al., 1993) for extensive-form games.1 At a self-confirming equilibrium, players maintain consistent beliefs of their opponents' strategies at information sets that are reached, but the beliefs of strategies can be incorrect at unreached information sets. Similarly, in our model, a fixed point can be different from a complete information equilibrium due to the incorrect estimates on the unobserved game outcomes formed by the beliefs (i.e., the estimated payoff distributions of strategies that differ from the fixed point may not be consistent). In general, these incorrect estimates may never be corrected by the learning dynamics because information of game outcomes is endogenously acquired based on the chosen strategies in each stage"

".This also allows us to show that the belief concentrates exponentially fast on the subset of parameters with the property that, at fixed point, each parameter in this set induces the same payoff distribution as the true parameter."

(Wu et al., 2022)

Convergence and Stability of Coupled Belief-Strategy Learning Dynamics in Continuous Games

"The phenomenon that endogenous information acquisition leads to incomplete learning is central to a variety of settings that include multi-arm bandit problems (Auer et al., 2002), Cesa-Bianchi and Lugosi [2006], Lattimore and Szepesvári [2020]), endogenous social learning (Banerjee, 1992), Gale and Kariv [2003], (Golub et al., 2010), (173028), (Mossel et al., 2012)), and self-confirming equilibrium learning in extensive form games (Fudenberg et al., 1993), (Fudenberg et al., 1995))."

(Dekel et al., 1999)

Payoff Information and Self-Confirming Equilibrium

"where players have no a priori information about opponents' play or payoffs, and each time the game is played they observe only the actions played by their opponents"

". We should make clear from the outset that, although this paper is motivated by the learning-theoretic approach to equilibrium in games, we do not here provide an explicit learning-theoretic foundation for our concepts. We are confident that such foundations can be constructed by, for example, incorporating restrictions on the priors into the steady-state learning model of Fudenberg and Levine [13], but we have not checked the details."

(Fudenberg et al., 2017)

Payoff information and learning in signaling games

"This paper is most closely related to the work of (Fudenberg et al., 1993), (Fudenberg et al., 2006), and (Fudenberg et al., 2017) on patient learning by Bayesian agents who believe they face a steady-state distribution of play"

".Our paper is also related to other models of Bayesian non-equilibrium learning, such as (Kalai et al., 1993) and (Esponda et al., 2014), though these papers do not study optimal experimentation and do not refine the Nash equilibrium set. Finally, Dekel et al. (1999), Fudenberg and Kamada (2015), and Fudenberg and Kamada (2018) develop equilibrium refinements that combine the idea of equilibrium arising from learning with the assumption that players know one another's payoff functions and feedback structures."

(Fudenberg et al._1, 2017)

Learning and Type Compatibility in Signaling Games

"the δ-stable outcomes are a version of self-confirming equilibrium where different types of sender are allowed to have different beliefs. Moreover, as the next example shows, this sort of heterogeneity in the senders' beliefs about the aggregate strategy of the receivers can endogenously arise in a δ-stable strategy profile even when all types of new senders start with the same prior over how the receivers play. 19"

"19 Dekel, Fudenberg, and Levine (2004) defined type-heterogeneous self-confirming equilibrium in static Bayesian games. As they noted, this sort of heterogeneity is natural when the type of each agent is fixed, but not if each agent's type is drawn i.i.d. in each period. To extend their definition to signaling games, we can define the "signal functions" y i (a, θ) from that paper to respect the extensive form of the game. See also ?."

(Fudenberg et al., 2006)

Superstition and Rational Learning

"We argue that some but not all superstitions can persist when learning is rational and players are patient, and illustrate our argument with an example inspired by the code of Hammurabi. The code specified an "appeal by surviving in the river" as a way of deciding whether an accusation was true, so it seems to have relied on the superstition that the guilty are more likely to drown than the innocent. If people can be easily persuaded to hold this superstitious belief, why not the superstitious belief that the guilty will be struck dead by lightning? We argue that the former can persist but the latter cannot by giving a partial characterization of the outcomes that arise as the limit of steady states with rational learning as players become more patient. These "subgame-confirmed

Nash equilibria" have self-confirming beliefs at information sets reachable by a single deviation. According to this theory a mechanism that uses superstitions two or more steps off the equilibrium path, such as "appeal by surviving in the river," is more likely to persist than a superstition where the false beliefs are only one step off of the equilibrium path."

Incomplete information and observational equivalence

With incomplete information, self-confirming equilibrium becomes especially permissive because many hidden states, type distributions, or opponent strategies can produce the same observations. In that case, learning can pin down only an observationally equivalent class of models, not the full underlying game, so SCE can differ sharply from Nash. (8 sources)

In incomplete-information settings, the core SCE idea is that beliefs only need to match what the player can actually infer from the feedback generated by play. A useful way later authors state this is that, by the end of the game, a player can be certain only of the **observationally equivalent** behavioral strategies of opponents and nature that are consistent with the realized path, and beliefs along reached information sets remain consistent with what is observed on that path. This makes the concept a coarsening of Nash for exactly the reason relevant here: the data generated by equilibrium play may identify only an equivalence class of hidden models, not the true one. (Schipper, 2017)

This is why observational equivalence is so central to Fudenberg-Levine style learning. If two different conjectures induce the same observable consequences, then repeated play does not tell the agent which one is right. Later work describes this as a general self-confirming logic: a strategy can be sustained by at least one belief compatible with observed behavior, and the concept can therefore be permissive when many underlying type distributions are observationally equivalent under the same observed mechanism or action pattern. (Feng et al., 2026)

The same point appears in more abstract statements about the relation between SCE and Nash. Recent work notes that self-confirming, conjectural, and subjective equilibrium all share the structure of **confirmed beliefs plus best reply**, and explicitly remarks that there are sufficient conditions under which Nash and self-confirming equilibrium are realization-equivalent. The flip side is important: absent those conditions, realization equivalence can fail because different hidden states or beliefs support the same realized play. (Battigalli et al., 2022) (Fudenberg et al., 1993)

In extensive-form or feedback-rich environments, what matters is not just actions but also **what information terminal outcomes reveal**. One way to formalize this is through terminal-node partitions: the analyst asks which terminal histories the player can distinguish after the game ends, and then studies equilibria relative to that feedback structure. Later papers show that an extensive-form game can be represented by a normal form plus these terminal-node partitions, preserving the same informational content; this helps explain why observational equivalence is not an afterthought but part of the definition of what players can learn. (Clark et al., 2022) (Fudenberg et al., 2015)

Incomplete information also opens the door to **type-dependent belief heterogeneity**. Dekel, Fudenberg, and Levine's extension to type-heterogeneous SCE allows different types to hold different beliefs that are each confirmed by their own experiences. Later discussions stress that this is natural when types are fixed: one type may systematically avoid an action and so never discover that another action would have been optimal given others' actual behavior. As a result, type-heterogeneous SCE can sustain outcomes that cannot arise in Nash equilibrium. (Fudenberg et al._1, 2017) (Fudenberg et al., 2019)

So for the user's themes, the main lesson is simple: with incomplete information, self-confirming equilibrium is driven by **what players can distinguish from what they observe**. When multiple latent structures are observationally equivalent on the realized path, learning need not recover the true game, and equilibrium can stop at self-confirmation rather than full Nash consistency. (Schipper, 2017) (Battigalli et al., 2022) (Fudenberg et al., 1993)

Evidence

(Schipper, 2017)

Self-confirming Games: Unawareness, Discovery, and Equilibrium

"Condition (ii) consists of two properties. First, at the end of the game the player is certain of observationally equivalent behavioral strategies of opponents and nature that allow her to reach the particular end of the game. That is, terminal beliefs are consistent with what has been observed during play (and hence at the end of the play). Second, beliefs do not change during the play. That is, beliefs at any information set reached during the play are consistent with what is observed at any point during the play and in particular with what is observed at the end of the game. Again, the idea is that everything that could have been learned on this path has been learned already in the past"

". While it is well-known in the literature (see for instance Fudenberg and Levine, 1993, Fudenberg and Kreps, 1995, Battigalli and Guaitoli, 1997) that self-confirming equilibrium is a coarsening of Nash equilibrium, the point here is to contrast it with the case of unawareness."

(Feng et al., 2026)

Self-Confirming Mechanisms

"We instead take a Bayesian approach. We say that a mechanism is an equilibrium if it is optimal for the designer under some belief in this set. This corresponds to the notion of selfconfirming equilibrium: the mechanism is sustained by at least one belief compatible with observed behavior.

A fourth, and more substantive, issue is that the equilibrium concept can be per-missive, since a wide range of mechanisms may be self-confirming given the many type distributions that are observationally equivalent under these mechanisms."

(Battigalli et al., 2022)

A framework for the analysis of self-confirming policies

"Indeed, conceptually this is a special case of the self-confirming equilibrium idea of Battigalli (1987) and Fudenberg and Levine (1993a, b), applied to one-person games with incomplete information about the probabilities of states"

".In the game-theoretic literature, a strategy profile that satisfies the properties of confirmed beliefs and subjective best reply has been called "conjectural equilibrium" (Battigalli, 1987;Battigalli & Guaitoli, 1988), "self-confirming equilibrium" (Fudenberg et al., 1993) and "subjective equilibrium" (Kalai et al., 1993)(Kalai et al., 1993)"

".Battigalli (1987) and (Fudenberg et al., 1993) provide sufficient conditions for the realizationequivalence between Nash and self-confirming equilibrium."

(Fudenberg et al., 1993)

Self-confirming equilibrium

"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts."

(Clark et al., 2022)

Observability, dominance, and induction in learning models

"To capture what is essential for learning outcomes in the normal form, we augment it with terminal node partitions (Fudenberg et al., 2015), Fudenberg and Kamada (2018)) which describe the information players observe when the game is played. We show that playing the extensive form game is equivalent to playing the normal form with the terminal node partitions that gives players the same information as would be revealed by the terminal nodes in the extensive form, so that the two games are information invariant."

(Fudenberg et al., 2015)

Rationalizable Partition-Confirmed Equilibrium

"Self-confirming equilibrium (SCE) formalizes the idea that incorrect off-path beliefs can persist for settings where players observe the terminal node of the game each time it is played, and the only restrictions placed on the players' beliefs is that they be consistent with the equilibrium distribution on terminal nodes. However, because SCE places no a priori restrictions on the players' beliefs, it does not capture the idea that players use prior information about opponents' payoff functions to predict the opponents' play. To capture such predictions, (Dekel et al., 1999) (hereafter DFL) define "rationalizable self-confirming equilibrium" (RSCE), which requires that players make certain inferences based on their knowledge of the other players' payoff functions and observation

structure."

(Fudenberg et al._1, 2017)

Learning and Type Compatibility in Signaling Games

"the δ-stable outcomes are a version of self-confirming equilibrium where different types of sender are allowed to have different beliefs. Moreover, as the next example shows, this sort of heterogeneity in the senders' beliefs about the aggregate strategy of the receivers can endogenously arise in a δ-stable strategy profile even when all types of new senders start with the same prior over how the receivers play. 19"

"19 Dekel, Fudenberg, and Levine (2004) defined type-heterogeneous self-confirming equilibrium in static Bayesian games. As they noted, this sort of heterogeneity is natural when the type of each agent is fixed, but not if each agent's type is drawn i.i.d. in each period. To extend their definition to signaling games, we can define the "signal functions" y i (a, θ) from that paper to respect the extensive form of the game. See also ?."

(Fudenberg et al., 2019)

Learning Theory and Heterogeneous Play in a Signaling-Game Experiment

"Dekel, Fudenberg, and Levine (2004) extended heterogeneous SCE to " type-heterogeneous self-confirming equilibrium" (or type-heterogenous SCE) to model heterogeneous beliefs in incomplete-information games. 3 With fixed types, it is thus possible that a Blue sender who selects Out never learns that receivers are selecting an action that makes it optimal to choose In . In fact, the game is constructed so that there is a type-heterogeneous SCE in which Blue senders select Out while Red senders select In , an outcome that cannot occur in a NE."

Relationship to Nash equilibrium and refinements

Self-confirming equilibrium is weaker than Nash because beliefs only have to be right where play is actually observed. Refinements try to shrink this gap by adding discipline on what off-path beliefs can survive, often using players' prior knowledge of payoffs, rationality, and the feedback structure. (4 sources)

The cleanest way to see the relation is this: every Nash equilibrium satisfies the SCE idea of best response plus correct on-path beliefs, but the converse can fail because SCE allows mistaken beliefs about what would happen after deviations or at unreached information sets. That is exactly why the concepts can come apart in dynamic games and other settings with limited feedback. At the same time, the gap is not always large: some later summaries note that SCE can be outcome-equivalent to Nash in two-player games even when it is not literally the same set of strategy profiles. (Fudenberg et al., 2009)

This relation becomes sharper in extensive-form settings. SCE was designed to capture cases where players repeatedly observe only terminal outcomes, so beliefs need only fit the induced distribution over terminal nodes. But once one wants a stronger link to strategic reasoning off the path, SCE can look too permissive, because it puts no prior restriction on beliefs that never get tested by play. That is one reason the literature compares it to more demanding dynamic concepts that are closer to sequential equilibrium, while relaxing full consistency on zero-probability paths in some classes of multi-stage or signaling environments. (Fudenberg et al., 2015) (Konicki et al., 2026)

The main refinement in the Fudenberg-Levine line is **rationalizable self-confirming equilibrium**. The idea is to keep the learning-based insight of SCE, but require players to use what they already know about opponents' payoff functions and rationality when forming beliefs. In this way, not every untested off-path story is allowed: some beliefs are ruled out because they are inconsistent with rational behavior given the known game. This refinement was introduced precisely to capture the fact that real players do not start from a blank slate; they often have structural knowledge that should shape prediction even before experience rules beliefs in or out. (Fudenberg et al., 2015) (Dekel et al., 1999)

So the big picture is that SCE sits between plain best-response behavior and full Nash or sequential discipline. It keeps the strong part of equilibrium that matters for observed play, but drops the requirement that beliefs be correct everywhere. Refinements such as RSCE try to recover some of that lost discipline without giving up the core learning logic that made self-confirmation attractive in the first place. (Fudenberg et al., 2015) (Dekel et al., 1999)

Evidence

(Fudenberg et al., 2009)

Learning and Equilibrium

> *"Self-confirming equilibrium (SCE): strategy profile in which each player's action is a best response to their beliefs about opponents' play, and each player's beliefs are correct along the equilibrium path are not Nash equilibria, as shown by (Fudenberg et al., 2010), but it is outcomeequivalent to Nash equilibrium in two-player games (Battigalli et al., 1997)(Fudenberg et al., 2010)."*

(Fudenberg et al., 2015)

Rationalizable Partition-Confirmed Equilibrium

> *"Self-confirming equilibrium (SCE) formalizes the idea that incorrect off-path beliefs can persist for settings where players observe the terminal node of the game each time it is played, and the only restrictions placed on the players' beliefs is that they be consistent with the equilibrium distribution on terminal nodes. However, because SCE places no a priori restrictions on the players' beliefs, it does not capture the idea that players use prior information about opponents' payoff functions to predict the opponents' play. To capture such predictions, (Dekel et al., 1999) (hereafter DFL) define "rationalizable self-confirming equilibrium" (RSCE), which requires that players make certain inferences based on their knowledge of the other players' payoff functions and observation structure."*

(Konicki et al., 2026)

Computing Perfect Bayesian Equilibria, with Application to Empirical Game-Theoretic Analysis

> *"Fudenberg and Tirole (Okada, 2011) introduced a more restrictive solution concept for dynamic games of imperfect information that is close to sequential equilibrium. Their version does not include the condition of consistency that is imposed by sequential equilibrium upon paths that are taken through the game tree with zero probability. However, this solution concept was developed specifically for a certain class of games known by several names: multi-stage signaling games, multi-period games with observed actions, multi-period games of imperfect information, multi-period games with independent types (Fudenberg et al., 1981)[19] (Okada, 2011)."*

(Dekel et al., 1999)

Payoff Information and Self-Confirming Equilibrium

> *"where players have no a priori information about opponents' play or payoffs, and each time the game is played they observe only the actions played by their opponents"*

> *". We should make clear from the outset that, although this paper is motivated by the learning-theoretic approach to equilibrium in games, we do not here provide an explicit learning-theoretic foundation for our concepts. We are confident that such foundations can be constructed by, for example, incorporating restrictions on the priors into the steady-state learning model of Fudenberg and Levine [13], but we have not checked the details."*

Variants, extensions, and related concepts

The self-confirming idea has been extended in several directions, mainly by changing what players observe, allowing small errors in confirmation, or moving to settings with many players and anonymous interaction. Across these variants, the common core stays the same: players best respond to beliefs that the feedback they actually receive does not refute. (2 sources)

- **Early parallel origins.** The concept was introduced in closely related form by Battigalli and by Fudenberg and Levine, and later work treats these as the main starting points of the SCE literature. (Cerreia-Vioglio et al., 2020) (Fudenberg et al., 1993)
- **\varepsilon-confirmation and approximate versions.** One extension weakens exact confirmation and allows beliefs to be only approximately consistent with observed feedback. Cerreia-Vioglio et al. note that “\varepsilon-confirmation” was introduced by Battigalli and Kalai-Lehrer, which places SCE inside a broader family where exact observational fit can be relaxed. (Cerreia-Vioglio et al., 2020)
- **Feedback-based variants.** A natural way to vary SCE is to change the feedback structure rather than the best-response part. The same source shows that if one drops perfect feedback but keeps exact confirmation (\varepsilon=0), the resulting condition is “arguably the nonatomic anonymous games counterpart” of SCE, which makes clear that the idea extends beyond the original finite-player setting.

(Cerreia-Vioglio et al., 2020)

- **Anonymous and nonatomic games.** In large games where a player cares about aggregates rather than named opponents, self-confirmation can be defined relative to what aggregate outcomes the player observes. This is a useful extension for the user's learning theme, because observational equivalence can be even stronger when many different micro-level action profiles generate the same aggregate signal. (Cerreia-Vioglio et al., 2020) (Model-Generated)
- **Subjective and conjectural equilibrium as close relatives.** SCE belongs to a wider family of concepts in which players optimize against subjective models that need only be confirmed by experience. In that sense, "subjective equilibrium" and "conjectural equilibrium" are best seen as neighboring ideas that differ in details of what beliefs are about and how strongly those beliefs must match the true environment. (Fudenberg et al., 1993) (Model-Generated)
- **A common core across variants.** Even when the setting changes—finite games, anonymous games, exact versus approximate confirmation—the recurring structure is the same notable Fudenberg-Levine one: best response plus correctness on the realized path, without requiring correct beliefs at unobserved contingencies. (Cerreia-Vioglio et al., 2020) (Fudenberg et al., 1993)
- **Why these variants matter for incomplete information and observational equivalence.** These extensions all push in the same direction: when feedback is coarse, aggregate, noisy, or only approximately informative, players can learn at most the part of the environment that their observations identify. That is exactly the setting where observationally equivalent hidden structures can support stable play without full Nash-level correctness. (Cerreia-Vioglio et al., 2020) (Fudenberg et al., 1993)

Paper	Equilibrium Type	Scope Of Equilibrium	Resolution Of Conceptual Problems	Player Beliefs	Applicability Conditions	Comparison With Nash Equilibrium
Equilibria of nonatomic anonymous games	$ \varepsilon$-estimated equilibrium, self-confirming equilibrium, peer-confirming equilibrium, Berk--Nash equilibrium	Encompasses self-confirming, peer-confirming, and Berk--Nash equilibria.	Addresses measurability issues and exogenous dependence in Schmeidler's framework.	Beliefs can be ε-inaccurate, allowing for imperfect confirmation and misspecification.	Positive $ \varepsilon$, measurable subsets, nontrivial finitely additive component.	Encompasses recent equilibrium concepts, resolves pure-strategy Nash issues.

Paper	Equilibrium Type	Scope Of Equilibrium	Resolution Of Conceptual Problems	Player Beliefs	Applicability Conditions	Comparison With Nash Equilibrium
Self-confirming equilibrium	Self-confirming equilibrium and Nash equilibrium.	Self-confirming equilibria can differ from Nash equilibria; includes standard solution concepts.	Differences between self-confirming equilibria and Nash equilibria.	Beliefs are correct on-path; off-path beliefs need not be correct.	Conditions under which beliefs are correct along equilibrium path.	Characterizes differences and provides conditions for correspondence with Nash equilibria.

Evidence

(Cerreia-Vioglio et al., 2020)

Equilibria of nonatomic anonymous games

"As noted in the Introduction, self-confirming equilibria were introduced for games with finitely many players by Battigalli [6] and Fudenberg and Levine (Fudenberg et al., 1993), and also ε-confirmation was introduced by Battigalli [6] and Kalai and Lehrer [17] and [18]"

".Finally, if we remove perfect feedback but maintain ε = 0, (5) becomes"

"which is arguably the nonatomic anonymous games counterpart of the definition of self-confirming equilibrium (SCE)."

(Fudenberg et al., 1993)

Self-confirming equilibrium

"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts."